

# Chapter 2 Detailed Work Packages (Implementation Guide)

Conditional Sequential Mixture of Flows (CSMF) for Imaging Inverse Problems

**Plan start:** 01 Oct 2025    **First outcomes draft:** 31 Mar 2026

## Detailed Work Packages (Implementation Guide)

**Notation.** We solve inverse problems of the form  $\mathbf{y} = A\mathbf{x} + \epsilon$  (additive) or  $\mathbf{y} = A\mathbf{x} \odot \epsilon$  (multiplicative, e.g., speckle), with forward operator  $A$  (blur+downsample for SR; identity or small filters in log domain for SAR). We build a *conditional* mixture of normalizing flows

$$q_{\Phi}(\mathbf{x} | \mathbf{y}) = \sum_{k=1}^K w_k(\mathbf{y}; \psi) q_k(\mathbf{x} | \mathbf{y}; \phi_k),$$

where  $w(\mathbf{y}; \psi) \in \Delta^{K-1}$  is a data-dependent gate, and each expert  $q_k$  is a flow conditioned on features  $c_{\eta}(\mathbf{y})$ .

### WP0 — Scope, infrastructure, baselines (01–15 Oct 2025)

**W0.1 Conditional experts (what “conditioning” means).** Each coupling/autoregressive block receives *features*  $h = c_{\eta}(\mathbf{y})$  from a conditioning network (MNIST: 4–6 layer CNN; SR/SAR: light UNet/ResNet encoder). We inject  $h$  via:

- **FiLM** (feature-wise linear modulation): for any intermediate feature  $f$ , apply  $\text{FiLM}(f, h) = \gamma(h) \odot f + \beta(h)$ , where  $\gamma, \beta$  are small MLPs over  $h$ .
- **Affine coupling nets:** scale/shift nets become  $s(\cdot; h)$  and  $t(\cdot; h)$  by concatenating  $h$  at input (or adding FiLM in hidden layers).

*Practical:* keep  $c_{\eta}(\mathbf{y})$  small (e.g., 64–128 channels at most); cache  $h$  per mini-batch to avoid recomputation across expert layers.

**W0.2 Global gating (what it is).** A gate  $g_{\psi}(\mathbf{y})$  outputs logits in  $\mathbb{R}^K$ ; weights are

$$w_k(\mathbf{y}; \psi) = \frac{\exp((g_{\psi}(\mathbf{y}))_k / \tau)}{\sum_j \exp((g_{\psi}(\mathbf{y}))_j / \tau)},$$

with temperature  $\tau \approx 1.0$ – $1.2$ . Start with a global vector  $w(\mathbf{y})$  (same for all pixels); later, extend to local (per-patch) maps.

*Minimal gate:* MNIST:  $\text{MLP}(\text{CNN}(\mathbf{y})) \rightarrow \mathbb{R}^K$ . SR/SAR: global average pooling on encoder features + 2 FC layers.

**W0.3 Baselines.** (i) Single conditional flow (one expert). (ii) Conditional mixture with *uniform* weights. Ensure same data preprocessing (normalization, crops) across models.

### W0.4 Unit tests.

1. Invertibility:  $\max \|\mathbf{x} - f^{-1}(f(\mathbf{x}))\|_{\infty} < 10^{-5}$ .
2. Log-det consistency: numerical Jacobian vs. sum of per-layer contributions on toy inputs.
3. Conditioning determinism: different  $\mathbf{y}$  produce different  $h$  and thus different  $s, t$ .

## WP1 — Measurement-consistency layer (16 Oct–15 Nov 2025)

**W1.1 Proximal/data-consistency step.** Insert a projection between flow blocks to reduce the residual to measurements.

$$\mathbf{x}^{\text{new}} = \arg \min_{\mathbf{z}} \underbrace{\frac{1}{\sigma^2} \|\mathbf{A}\mathbf{z} - \mathbf{y}\|_2^2}_{\text{data fidelity}} + \underbrace{\lambda \|\mathbf{z} - \mathbf{x}\|_2^2}_{\text{stay close}}, \Rightarrow \quad (1)$$

$$\mathbf{x}^{\text{new}} = \left( \frac{1}{\sigma^2} \mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I} \right)^{-1} \left( \frac{1}{\sigma^2} \mathbf{A}^\top \mathbf{y} + \lambda \mathbf{x} \right). \quad (2)$$

*SAR (blur+downsample)*: let  $\mathbf{A} = \mathbf{D}\mathbf{B}$  (downsample after blur). Use adjoints:  $\mathbf{B}^\top$  as flipped kernel;  $\mathbf{D}^\top$  as nearest/zero-upsample. Approximate the inverse by one *preconditioned conjugate gradient* (PCG) step or a closed form in Fourier if  $\mathbf{D} = \mathbf{I}$ :

$$\hat{\mathbf{x}}^{\text{new}}(\omega) = \frac{\overline{\hat{\mathbf{B}}(\omega)} \hat{\mathbf{y}}(\omega) / \sigma^2 + \lambda \hat{\mathbf{x}}(\omega)}{|\hat{\mathbf{B}}(\omega)|^2 / \sigma^2 + \lambda}.$$

*SAR (multiplicative noise)*: work in log domain:  $\tilde{\mathbf{y}} = \log \mathbf{y}$ ,  $\tilde{\mathbf{x}} = \log \mathbf{x}$ . Consistency loss becomes  $\|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|^2$  (optionally with local variance weights).

**W1.2 Plug-and-Play sampling.** At generation time, alternate a few steps (flow  $\rightarrow$  prox):

$$\mathbf{x}^{(0)} \sim q_\Phi(\cdot \mid \mathbf{y}), \quad \mathbf{x}^{(t+1)} = \text{Prox}_A(\mathbf{x}^{(t)}), \quad t = 0, \dots, T-1.$$

Typically  $T = 1\text{--}3$  is enough; monitor  $\|\mathbf{A}\mathbf{x}^{(t)} - \mathbf{y}\|$ .

## WP2 — Hybrid objective & schedule (16 Nov–15 Dec 2025)

### W2.1 Loss definition.

$$\mathcal{L} = \underbrace{-\mathbb{E}_{\mathbf{y}} \mathbb{E}_{\mathbf{x} \sim q_{\Phi}(\cdot | \mathbf{y})} [\log p_{\theta}(\mathbf{x} | \mathbf{y})]}_{\text{NLL/ELBO}} + \lambda_{\text{cons}} \mathbb{E}_{\mathbf{y}, \mathbf{x}} [d(A\mathbf{x}, \mathbf{y})] + \lambda_{\text{trans}} \mathcal{T}(\mathbf{x}; \mathbf{y}) + \lambda_{\text{cal}} \mathcal{C}(\mathbf{x}; \mathbf{y}), \quad (3)$$

with  $d(\cdot)$  an  $L_2$  or noise-aware Mahalanobis residual;  $\mathcal{T}$  a geometry term (differentiable sliced-Wasserstein or MMD); and  $\mathcal{C}$  a calibration term (CRPS or temperature penalty).

**W2.2 Sliced-Wasserstein (differentiable).** Sample  $L$  random directions  $\{\mathbf{u}_{\ell}\}_{\ell=1}^L$  on the unit circle; project samples  $\alpha = \mathbf{u}_{\ell}^{\top} \mathbf{x}$ , and compare empirical 1D distributions by sorting:

$$\text{SW}_2^2 \approx \frac{1}{L} \sum_{\ell=1}^L \frac{1}{M} \sum_{i=1}^M (\alpha_{\ell}^{(i)} - \tilde{\alpha}_{\ell}^{(i)})^2,$$

where  $(\cdot)^{(i)}$  are sorted order statistics from model vs. reference/target. Use  $L = 256$ – $512$ ; gradients propagate through sorting (subgradients suffice in practice).

**W2.3 MMD (multi-bandwidth).** Use a mixture of RBF kernels  $k_{\sigma}(\mathbf{x}, \mathbf{x}') = \exp(-\|\mathbf{x} - \mathbf{x}'\|^2 / 2\sigma^2)$  with  $\sigma \in \{\sigma_1, \dots, \sigma_B\}$  determined by the median heuristic on the batch; average MMD across bandwidths for stability.

**W2.4 Calibration term.** For high-dimensional images, prefer the *energy score* (proper scoring rule):  $\text{ES} = \mathbb{E}\|\mathbf{x} - \mathbf{x}'\| - 2\mathbb{E}\|\mathbf{x} - \mathbf{x}^{\star}\|$ , where  $\mathbf{x}, \mathbf{x}'$  are two samples from  $q_{\Phi}(\cdot | \mathbf{y})$  and  $\mathbf{x}^{\star}$  a (pseudo-)reference. Alternatively, apply *temperature scaling* on latent/base scales and penalize deviation from nominal coverage on validation.

### W2.5 Training schedule.

1. **Stage A (experts only):** train each  $q_k(\mathbf{x} | \mathbf{y})$  with NLL + weak consistency ( $\lambda_{\text{cons}} \in [0.01, 0.05]$ ), gate frozen to uniform; early stop on NLL.
2. **Stage B (gate only):** freeze experts; train  $w_{\psi}(\mathbf{y})$  with the full hybrid loss; use temperature  $\tau \approx 1.1$  and optional top- $k$  masking ( $k = 2$ – $3$ ).
3. **Stage C (light joint):** unfreeze last 1–2 blocks per expert and the gate; small LR; keep consistency active; early stop on validation NLL+residual.

### WP3 — Ablations on MNIST (16 Dec 2025–20 Jan 2026)

**W3.1 Gate ablations.** *Global vs. local:* local gate produces  $W(\mathbf{y}) \in \mathbb{R}^{K \times H \times W}$  (softmax over  $K$  per spatial location) and fuses expert outputs pixelwise:  $\hat{\mathbf{x}} = \sum_k W_k(\mathbf{y}) \odot \hat{\mathbf{x}}_k$ . Start with *block-wise* gates (e.g.,  $8 \times 8$  tiles with shared weight per tile) to save memory. *Top- $k$ :* zero all but the best  $k$  logits before softmax; stabilize with  $\tau \in [1.0, 1.3]$ .

**W3.2 Consistency ablations.** Number of prox steps  $T \in \{0, 1, 2, 3\}$ ; proximal weight  $\lambda \in \{0.05, 0.1, 0.2\}$ ; PCG iters  $\in \{1, 3\}$ . Track  $\|A\mathbf{x} - \mathbf{y}\|$ , PSNR/SSIM, and NLL trade-off.

**W3.3 Geometry & calibration.** SW<sub>2</sub>:  $L \in \{128, 256, 512\}$ . MMD:  $B \in \{3, 5\}$  bandwidths (log-spaced around median). Calibration: ES vs. CRPS; with/without temperature scaling.

**W3.4 Reporting.** Pareto curves of NLL vs.  $\|A\mathbf{x} - \mathbf{y}\|$ ; SW<sub>2</sub>/MMD vs. NLL; coverage/PIT histograms. Include gate usage statistics (effective #experts  $N_{\text{eff}} = \exp(H(w))$ ).

## WP4 — Port to imaging tasks (21 Jan–10 Mar 2026)

### W4.1 Optical SR pipeline.

1. **Data:** extract grayscale patches (e.g.,  $64 \times 64$  HR) from DIV2K/BSD; synthesize LR via blur (Gaussian with  $\sigma \in [0.8, 1.8]$ ) and downsample ( $\times 2$ ,  $\times 4$ ), add AWGN with known  $\sigma$ .
2. **Conditioner**  $c_\eta$ : light UNet encoder producing  $h \in \mathbb{R}^{C \times H' \times W'}$  ( $C = 64$ ); global pooled  $h$  for global gate; optional local gate on  $H' \times W'$  map.
3. **Consistency:** PCG prox with adjoints  $B^\top, D^\top$ ; one step per block or after  $L$  blocks.
4. **Metrics:** PSNR/SSIM on reconstructions; residual  $\|A\mathbf{x} - \mathbf{y}\|$ ;  $\text{SW}_2/\text{MMD}$  between posterior samples and reference HR patch distribution; ES/coverage on synthetic pairs.

### W4.2 SAR despeckling (prototype).

1. **Stabilization:** work with  $\tilde{\mathbf{y}} = \log \mathbf{y}$ ; model  $\tilde{\mathbf{x}}$ ; consistency is  $\|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|^2$  (or weighted by local variance).
2. **Conditioner:** CNN encoder on  $\tilde{\mathbf{y}}$ ; same gating options as SR.
3. **Metrics:** ENL in homogeneous regions, edge preservation index, radiometric bias, residual to multi-look reference (if available), ES.

**W4.3 Practical defaults.**  $K = 3$  experts; gate  $\tau = 1.1$ ; top- $k = 2$ ; prox  $T = 1$  per sampling;  $\lambda_{\text{cons}} = 0.1$ ,  $\lambda_{\text{trans}} = 0.05$  (anneal).

## WP5 — First outcomes & write-up (11–31 Mar 2026)

**W5.1 Freezing & sweeps.** Fix seeds/configs; run final sweeps over  $\lambda_{\text{cons}}$ ,  $T$ , and gate  $\tau$ . Save all ablation tables and plots with commit hashes.

### W5.2 Figures & tables.

- Qualitative: posterior sample grids (with/without prox), SR reconstructions ( $\times 2$ ,  $\times 4$ ), SAR despeckled examples.
- Quantitative: NLL vs. residual;  $\text{SW}_2$ /MMD; PSNR/SSIM (SR), ENL/bias (SAR); calibration (PIT/ES).
- Gate diagnostics: histogram of  $w(\mathbf{y})$ , spatial maps for local gate,  $N_{\text{eff}}$  trajectories.

**W5.3 Writing checklist.** Method (CSMF), conditioning/gating design, consistency layer, hybrid objective, training schedule, MNIST ablations, SR results, proto-SAR, limitations (gate collapse risk; prox–NLL trade-off), and reproducibility notes (data, seeds, configs).

## Appendix: Minimal code templates (language-agnostic)

### A1. FiLM in a coupling block.

```
# inputs: x = [xA, xB], features h = c(y)
gamma, beta = MLP_gamma(h), MLP_beta(h)
u = CouplingNet(xA)
u = gamma * u + beta
s, t = Head(u).chunk(2, dim=1)
xB_out = xB * exp(tanh(s) * s_max) + t
```

# FiLM params  
# hidden  
# FiLM modulate  
# scale, shift

### A2. Global gate with top-k.

```
logits = GateNet(h_global)
if topk < K:
    keep = logits.topk(k=topk).indices
    mask = full_like(logits, -inf); mask[keep] = 0
    logits = logits + mask
w = softmax(logits / tau, dim=-1)
```

#  $R^K$

### A3. One-step SR prox (Fourier, no downsample).

```
X = FFT(x); Y = FFT(y); H = FFT(kernel)
num = conj(H) * Y / sigma2 + lam * X
den = (abs(H)**2) / sigma2 + lam
X_new = num / (den + 1e-8)
x_new = IFFT(X_new).real
```

### A4. Sliced-Wasserstein (batch).

```
directions = randn(L, d); directions /= norm(directions, dim=1, keepdim=True)
alpha = X @ directions.T # (B, L)
alphaT = X_ref @ directions.T # (B, L)
alpha_sorted = sort(alpha, dim=0).values
alphaT_sorted = sort(alphaT, dim=0).values
sw2 = mean((alpha_sorted - alphaT_sorted)**2)
```

### A5. Energy score (approx).

```
# X: S samples from q(x|y), X': another S samples, x*: reference or held-out
es = mean(norm(X - X_prime, dim=1)) - 2 * mean(norm(X - x_star, dim=1))
```